

# Transformations

A regression is fitted on a scale, and nothing in least squares says which one. Food expenditure can be modelled in currency units or in logarithms, a count as it stands or through its square root. The choice affects all three claims of the normal linear model: a linear mean, a constant variance and normal errors. When the three failures share a cause, one transformation repairs them together; when they do not, the chapter says which to fix first and at what price. It treats the choice of scale as estimation: the Box–Cox profile likelihood for the response, the Box–Tidwell procedure and component-plus-residual plots for regressors, and the delta method for transformations that stabilize a variance tied to the mean. It ends with what a transformed model says on the scale the reader cares about: medians, means, prediction intervals and elasticities.

**What you need.** Least squares as projection and the Frisch–Waugh–Lovell theorem ([Chapter 6](#), Theorem 6.6.2), maximum likelihood and profile likelihood in the normal linear model ([Chapter 7](#), Theorem 7.3.1 and Definition 7.3.1), the interpretation of log-scale coefficients ([Section 5.7](#)), prediction intervals (Theorem 12.3.1), the delta method as used in [Section 14.4](#), and the diagnostics of [Chapter 20](#) and [Chapter 21](#).

## 22.-1.1 Roadmap

- [Transforming the response](#)
- [The Box–Cox family](#)
- [Transforming regressors](#)
- [Variance-stabilizing transformations](#)
- [Interpreting a transformed model](#)
- [Summary and notes](#)